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MBDA: A modality-balanced framework with data augmentation and alignment for multimodal emotion recognition

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ABSTRACT

Multimodal Emotion Recognition (MER) aims to infer human emotional states by integrating complementary information from heterogeneous modalities. However, existing MER methods often suffer from modality imbalance, cross-modal misalignment, and limited data diversity, which hinder their robustness and generalization. To address these issues, we propose a Modality-Balanced framework with Data Augmentation and Alignment (MBDA), which integrates modality-aware augmentation, feature alignment, and counterfactual knowledge distillation into a unified framework in a progressive learning manner. MBDA boosts data diversity while preserving semantic consistency through modality-aware augmentation, enforces robust multi-level alignment across modalities, and adaptively rebalances modality contributions through counterfactual knowledge distillation. Experiments on the DEAP and SEED-IV datasets demonstrate that MBDA consistently outperforms state-of-the-art methods, achieving accuracies of 93.86%, 95.11%, 91.02%, and 92.66% on DEAP-A, DEAP-V, DEAP-AV, and SEED-IV, respectively.

1. Introduction

Emotions are integral to human communication, exerting a profound influence on how we think, behave, and connect with others in daily life. However, given the complexity of human emotional expression, single-modality approaches are often insufficient. Multimodal Emotion Recognition (MER), which integrates data such as speech, facial expressions, and physiological signals, provides a more holistic and reliable understanding of human emotions (Wu et al., 2025), and has thus become a key direction in affective computing (AC) and human-computer interaction (HCI).

Among these modalities, physiological signals, such as heart rate, skin conductance, and electroencephalogram (EEG) data, can directly reflect neural activity. They offer objective and sensitive indicators of emotional arousal with strong resistance to external interference, making them well-suited for capturing authentic emotional states (Lian et al., 2023). Nevertheless, their reliance on specialized acquisition equipment and lack of clear situational or semantic interpretability pose limitations. By contrast, non-physiological modalities, such as facial expressions and eye movements, are intuitive and rich in contextual information but are more sensitive to environmental factors, cultural differences, and in-

dividual behavior, reducing their stability. Combining EEG with facial expressions or eye movements allows each modality to complement the weaknesses of the other, with EEG providing robust neural evidence and facial or eye-movement signals offering intuitive and semantically meaningful cues. Therefore, this multimodal fusion enables a more comprehensive and reliable representation of emotional states (Vairamani, 2024), making it particularly valuable for advancing MER.

However, due to the differences among modalities in data quality, signal type, and noise robustness, models are prone to modality imbalance during training. Specifically, dominant modalities with higher quality and stronger robustness tend to prevail in prediction, while weaker modalities are often neglected due to insufficient information or vulnerability to interference. In such cases, a dominant modality tends to prevail, while comparatively weaker modalities are often overlooked. To mitigate this problem, numerous balancing strategies have been developed, including sample reorganization-based data mixing (Ma et al., 2025), the dynamically weighted Ada2I model (Nguyen et al., 2024), and continual enhancement mechanisms (Jiang et al., 2025). Nevertheless, most of these methods rely on static or predefined rules, lacking the ability for context-aware dynamic adaptation. As a result, weaker modalities can be overshadowed by dominant ones, leading the model

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to neglect their information and over-rely on the dominant modality, which reduces overall robustness. Therefore, how to construct a dynamically adaptive modality balancing mechanism remains a critical challenge in current MER research.

Secondly, MER often faces challenges of sample scarcity and noise interference, which undermine both the stability of model training and the reliability of prediction. To address these issues, various data augmentation strategies have been proposed. For example, generative adversarial networks (GANs) (Liu et al., 2023), mix-based augmentation (Cai et al., 2024), and diffusion models (Siddhad et al., 2024) can expand training datasets, enrich minority-class samples, and enhance model adaptability. While such approaches alleviate data imbalance and noise to some extent, they also introduce new challenges. Excessive or inconsistent augmentations may disrupt intra-modal semantic structure, cause feature space drift, and reduce training stability. Moreover, uneven augmentation across modalities can exacerbate cross-modal misalignment, weakening semantic consistency and ultimately degrading recognition performance. Therefore, how to enhance sample quality through augmentations of varying intensities while preserving semantic consistency within and across modalities remains a critical challenge in MER.

Additionally, feature alignment across modalities remains a key challenge in MER. Owing to the heterogeneous nature of multimodal signals, their distribution forms, semantic representations, and discriminative capabilities differ significantly. As a result, simple concatenation or dimensional mapping cannot fully exploit their complementary strengths. Without effective alignment, high-level representations may suffer from semantic mismatches, undermining fusion performance and degrading recognition accuracy. To tackle this problem, researchers have investigated different strategies such as cross-modal attention for explicit alignment (Chen & Zhang, 2024) and contrastive learning for implicit alignment (Zhao et al., 2024). Although these methods improve static alignment and semantic consistency, they often neglect representation consistency within the same modality under varying augmentation strengths, leaving models vulnerable to intra-modal drift. Therefore, achieving robust cross-modal semantic alignment while preserving intra-modal consistency under augmentation remains a critical challenge for advancing MER.

To address the challenges above, we introduce MBDA, which integrates modality-aware data augmentation, multi-level feature alignment, and counterfactual knowledge distillation as a unified framework. Specifically, modality-aware data augmentation is coupled with contrastive consistency learning to enhance sample diversity while preserving the semantic integrity of the original signals across different augmentations. Multi-level feature alignment enforces both intra-modal and inter-modal semantic consistency, ensuring reliable cross-modal interactions under varying augmentation intensities. Counterfactual knowledge distillation adaptively modulates the contributions of different modalities, enhancing weaker modalities while suppressing redundant information from dominant ones. These modules do not function in isolation, but rather interact in a closed-loop, causality-driven manner. The augmentation intensity influences the alignment objectives, the alignment consistency informs the distillation process, and the distillation module dynamically adjusts modality contributions during training. This integrated approach ensures that the three components mutually reinforce each other, achieving robust cross-modal semantic alignment, strengthening the representation of weaker modalities, and boosting overall emotion recognition performance. In contrast to traditional methods, MBDA establishes an interdependent optimization process rather than relying on a simple concatenation or modular stacking of independent components. This novel design fully leverages complementary information from different modalities while preserving intra-modal consistency, yielding a more effective and robust framework for MER.

The primary contributions of this study can be summarized as follows:

1. We propose a unified modality-balanced framework (MBDA) for MER, which jointly models multiple modalities, augmentation strengths, and semantic levels within a single learning paradigm. Unlike previous approaches, the framework leverages the interdependence of augmentation, alignment, and distillation, forming a closed-loop optimization process.
2. We introduce a multi-level and multi-view alignment mechanism that simultaneously enforces intra-modal consistency across varying augmentation strengths and inter-modal semantic alignment across heterogeneous modalities. This ensures that alignment is robust to augmentation-induced distribution shifts and maintains semantic coherence under realistic training perturbations.
3. We design a counterfactual knowledge distillation module that constructs cross-modal ranking relationships and counterfactual samples, adaptively adjusting modality contribution weights. By explicitly considering modality interactions, the framework enhances weak modalities, mitigates dominance bias, and further improves accuracy and robustness in MER.

2. Related work

2.1. Modality imbalancing for MER

In MER, modality imbalance causes dominant modalities to overshadow weaker ones, reducing the model's overall performance and robustness. To tackle this issue, researchers have developed various approaches aimed at reducing modality bias. For example, Nguyen et al. (2024) proposed Ada2I, which applies dual-layer adaptive balancing at the feature and modality levels with adaptive weighting and a Disparity Ratio metric to adjust weak modalities. Li et al. (2023b) designed Decoupled Multimodal Distillation, splitting each modality into shared and private subspaces to enable local compensation and coordination between dominant and weak modalities. Woo et al. (2025) developed MAAN, which integrates modal perception attention and agent-based multimodal loss to alleviate fusion insufficiency from modality imbalance. Wu et al. (Wu et al., 2025) proposed CDGKD, employing modality rebalancing and intensive guidance to transfer knowledge from multimodal teachers to unimodal students, improving visual-based emotion recognition. Wang et al. (2023) presented HCT-DMG, detecting inter-modal inconsistencies and applying dynamic gating to prioritize dominant modalities, reducing misleading effects from mode conflicts. Kim et al. (2024) developed CausalMER, leveraging causal graph construction and counterfactual reasoning to identify biased modality influences, achieving balanced contributions and robust performance.

While prior methods mitigate modality imbalance at different levels, most rely on static rules and lack context-aware adaptation, causing weak modalities to be overshadowed by dominant ones and reducing robustness. To address this, we propose a dynamic modality balancing mechanism based on counterfactual knowledge distillation. By constructing cross-modal counterfactual samples and ordering relationships, the module compares prediction differences between teacher and student models, guiding students to extract latent causal information, dynamically adjust modality weights, suppress redundancy of dominant modes, and enhance weak modality representations.

2.2. Data augment for MER

In MER, data scarcity and modal noise hinder performance improvements. To address these challenges, diverse data augmentation strategies have been explored. GAN-based methods include (Baek & Lee, 2023), who applied a DCGAN speech enhancement approach using mel spectra, and Roh et al. (Roh & Lee, 2024), who extended it with Conditional-DCGAN and reinforcement learning, both yielding improved recognition accuracy. For EEG, Bao et al. (2021) proposed

VAE-D2GAN to enhance sample diversity, achieving higher accuracy on SEED and SEED-IV. Diffusion-based approaches have also shown promise. For example, Siddhad et al. (2024) employed DDPM for EEG generation with significant gains on SEED, SEED-IV, and DEAP, while Roy et al. (2024) used Stable Diffusion for facial expression data augmentation, improving results on FER2013 and RAF-DB. Beyond generation, Cai et al. (2024) introduced Mixup in multimodal spaces with contrastive alignment to handle noisy inputs, and Jin et al. (2025) proposed RoHyDR, integrating diffusion and adversarial learning to recover missing modalities, thereby enhancing fusion robustness. And Wang et al. (2024) explored multimodal fusion under challenging conditions, such as low-light environments, providing insights that could inspire improved emotion recognition.

However, a key challenge in MER is improving sample quality through varying augmentation intensities while preserving semantic consistency across modalities. To address this, we propose an intensity-aware augmentation mechanism that, unlike static methods, leverages contrastive learning to enforce consistency within augmented views of the same modality. This design prevents semantic drift from over-optimization, thereby mitigating feature space shifts and cross-modal alignment deviations during augmentation.

2.3. Data alignment for MER

In the field of MER, achieving cross-modal semantic alignment while maintaining intra-modal consistency has long been a central challenge. Recent studies have explored diverse strategies to improve fusion effectiveness. For example, Wang et al. (2025) introduced a multi-granularity alignment framework across distribution, instance, and label levels. Li et al. (2024) proposed Foal-Net to integrate audio-visual features via an alignment module. Shi and Huang (2023) designed a sample-weighted focal contrastive loss to preserve intra-modal consistency and improve minority class recognition. Shou et al. (2023) enhanced feature robustness through graph-based contrastive learning. Xie et al. (2025) developed the DCLF framework with context-aware comparison and pseudo-labeling to mitigate dominance bias. Guo et al. (2023) introduced ME-MHACL with self-supervised contrastive learning and multi-head attention to improve cross-individual generalization. And Li et al. (2023a) proposed the Joyful framework, using joint modality fusion and graph contrastive learning to improve cross-modal alignment.

While these methods advance alignment techniques, most emphasize single-perspective alignment and overlook varying augmentation intensities. To bridge this gap, we develop an alignment module that jointly optimizes intra-modal alignment losses under varying augmentation intensities and inter-modal alignment losses under consistent augmentation. By enforcing multi-level and multi-perspective constraints, our approach achieves robust feature alignment at both semantic and modality levels, thereby enhancing cross-modal semantic consistency.

3. Method

The proposed MBDA model, as illustrated in Fig. 1. First, the model extracts shallow features from raw EEG and facial inputs. Then, these features are augmented with weak and strong strategies and processed by modality-specific encoders to obtain deep representations. Next, a modality-balancing stage with teacher-student distillation and counterfactual weighting ensures fair contributions, after which cross-modal alignment. Finally, fusion and classification are performed for emotion recognition.

3.1. Definition

Given the raw multimodal data of N samples, denoted as $\mathbf{X}_m \in \mathbf{R}^{N \times D_m}$, $m \in \{eeg, face, eye\}$, where N denotes the total number of sample, and D_m corresponds to the dimensionality of the original feature space for modality m . The ground-truth label set $\mathbf{Y}$ is defined as

$\mathbf{Y} = \{\mathbf{Y}_i\}_{i=1}^N$, $\mathbf{Y}_i \in \{1, 2, \dots, C\}$, where C represents the number of emotion category. The primary objective of this work is to accurately predict emotional categories by leveraging multimodal representations.

3.2. Modality-specific feature extract

Due to the heterogeneity of signal forms and distributions across modalities, we first extract initial features independently. In this study, modality-specific extractors map the input $\mathbf{X}_m$ represents the original feature dimension of modality m into shallow representations $\tilde{\mathbf{X}}_m \in \mathbf{R}^{N \times D}$. Here, the output dimensions of all modalities are unified to D , enabling alignment and facilitating subsequent encoding and cross-modal fusion.

3.2.1. EEG feature extraction

We first employ a multi-layer convolutional network to extract the spatial dependencies among electrodes. The resulting spatial features are then fed into a temporal convolution module to capture long-range temporal contexts and sequential dynamic patterns, yielding discriminative spatiotemporal representations. By reinforcing spatial activation patterns linked to emotional states, this fusion contributes to enhanced overall performance in emotion recognition. The entire process can be expressed as:

$$\mathbf{X}_{s,l}^e = G_{s,l}^e(\mathbf{X}_{s,l-1}^e) + \mathbf{X}_{s,l-1}^e, \quad l = 1, \dots, L \quad (1)$$

$$\tilde{\mathbf{X}}^e = \mathbf{X}_l^e = \text{TemporalConv}(\mathbf{X}_s^e) \quad (2)$$

where $\mathbf{X}_{s,l}^e$ denotes the spatial features at the l th convolutional layer, $\mathbf{X}_s^e$ denotes the final output of the spatial convolution block, $G_{s,l}^e(\cdot)$ represents the convolution operation, and $\tilde{\mathbf{X}}^e$ is the fused EEG feature embedding.

3.2.2. Face feature extraction

Facial expressions are complex signals that contain both rich spatial structures and local muscle movement patterns. In this work, we compute features from 16 landmark points to represent facial regions. The indices of these landmarks are 19, 22, 23, 26, 28, 31, 37, 38, 41, 45, 46, 48, 49, 54, 55, and 58. First, a convolutional encoder is employed to extract local appearance features. These features are then upsampled along the temporal dimension and mapped into EEG-aligned channels. Finally, a band mapping module further refines the features to ensure consistency with physiological modalities. The hierarchical design allows the model to effectively capture both fine-grained local features and the global spatial context of facial expressions. The process can be expressed as:

$$\mathbf{X}_c^f = \text{Linear}(\text{Upsample}(\text{Conv}(\mathbf{X}_f))) \quad (3)$$

$$\tilde{\mathbf{X}}^f = \text{BandMap}(\mathbf{X}_c^f) \quad (4)$$

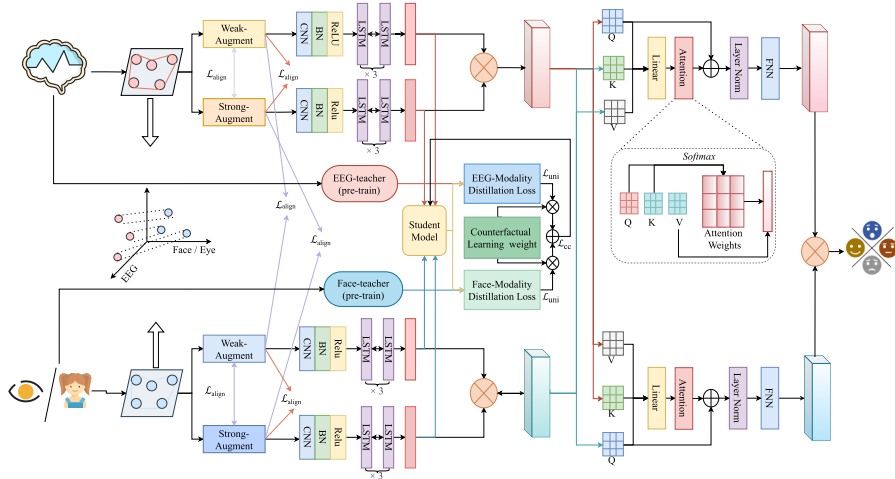
where $\mathbf{X}_f$ is the input facial feature, $\text{Upsample}(\cdot)$ matches the temporal length to EEG, $\text{Linear}(\cdot)$ maps the feature into EEG-aligned channels, and $\text{BandMap}(\cdot)$ enhances the representation across frequency bands.

3.2.3. Eye movements feature extraction

Eye movements reflect gaze patterns and attention shifts during emotional processing. In this study, we utilize the SEED dataset, where preprocessed eye-movement features are represented as 31-dimensional signals. We first apply a one-dimensional convolution to capture local dependencies within gaze sequences, then upsample the temporal length from 10 to 50 to match EEG signals. Finally, a channel mapping operation is performed to align the feature dimension with EEG channels, resulting in temporally synchronized eye-movement features. The process is defined as:

$$\mathbf{X}_s^o = \text{Conv}_{map}(\text{Upsample}(\text{Conv}(\mathbf{X}^o))) \quad (5)$$

where $\mathbf{X}^o$ is the raw eye-movement signal.



3.3. Data augment

3.4. Features encoder

To mitigate modality imbalance between physiological and non-physiological signals, our method is inspired by the modality-balanced learning framework proposed by Zhang et al. (2024). Specifically, we extend their idea of training unimodal teachers with the same backbone architecture as the multimodal student. This ensures a fair comparison without relying on stronger unimodal models. Different from the original application in multimedia recommendation, we adapt this paradigm to the emotion recognition task. By comparing predictions under counterfactual substitutions, the model learns which modality provides more informative cues for each instance. To mitigate modality imbalance, we propose to re-estimate the modality-specific encoders based on the distributional characteristics of their output feature spaces. Specifically, the counterfactual distillation process is realized by implicitly constructing cross-modal alternative pairs within each training batch. For a given sample, the student model is guided to compare its prediction with counterfactual predictions obtained by replacing one modality with the corresponding teacher representation of another modality. This design enables the model to reason about modality contributions through relative ranking differences rather than absolute output matching. This design enables the model to reason about modality contributions through relative ranking differences rather than absolute output matching, as illustrated in Algorithm 1.

Algorithm 1: Counterfactual distillation for modality-balanced emotion recognition.

Input: Multimodal dataset $D = \{(\mathbf{x}_e, \mathbf{x}_f, \mathbf{x}_o, y)\}$,
Unimodal teachers F_m^t , student f^s , temperature τ , epsilon ϵ

Output: Trained student model f^s

```

1 Pre-train teachers: for  $m \in \{eeg, face, eye\}$  do
2   Train  $F_m^t$  with modality- $m$  inputs, generate predictions (Eq.
   (15))
3 Initialize student:  $f^s$  with random weights, generate
   predictions (Eq. (16)) for each epoch do
4   for each batch  $B \subseteq D$  do
5     1: Compute ranking scores (Eq. (17))
6     2: Specific Distillation Loss  $\mathcal{L}_{SD}$  (Eq. (19))
7     3: Generic Distillation Loss  $\mathcal{L}_{GD}$  (Eq. (20))
8     4: Counterfactual Weighting  $\lambda_i^m$  (Eq. (23)–(29))
9     5: Update student with total distillation loss  $\mathcal{L}_{dis}$ 
10 return  $f^s$ 

```

3.5.1. Unimodal teacher and multimodal student

In our framework, both teacher and student models are designed with an identical backbone architecture. This ensures that the difference in their outputs arises solely from the input modalities, rather than from the model capacity. Although three modalities (EEG, facial expressions, and eye movements) are mentioned for completeness, in our experiments, we use only two modalities at a time, either EEG with facial expressions or EEG with eye movements, depending on the dataset.

- **Teacher models:** For each modality $m \in \{eeg, face, eye\}$, we train a unimodal teacher model F_m^t using only the corresponding modality input, while ablating other modalities. Specifically, the EEG teacher f_e^t receives only EEG signals, and the face teacher f_f^t receives only facial features, and the eye movement teacher f_o^t receives only facial features. Their predictions are formulated as:

$$\hat{y}^t = f^t(\mathbf{x}_e, \mathbf{x}_{f/o}) \quad (15)$$

where $\mathbf{x}_e$ denotes EEG inputs, and $\mathbf{x}_{f/o}$ denotes facial or eye movement inputs, respectively.

- **Student model:** The multimodal student f^s adopts the same backbone for each modality, followed by a cross-modal fusion layer to integrate representations. The student simultaneously processes both EEG and face inputs:

$$\hat{y}^s = f^s(\mathbf{x}_e, \mathbf{x}_f, \mathbf{x}_o) \quad (16)$$

3.5.2. Specific distillation loss

During training, we construct positive and negative pairs. In our counterfactual distillation framework, the construction of these pairs follows the strategy proposed in Zhang et al. (2024) for modality-balanced learning, where positive pairs correspond to the same instance in the same modality, and negative pairs are drawn from other instances, similar to the Bayesian Personalized Ranking (BPR) paradigm. We apply this strategy to the MER task to ensure semantic consistency and effective cross-modal guidance. For an anchor sample i , the positive sample represents the same instance in the same modality, ensuring semantic consistency. The negative sample is defined as the representation of a different instance $j \neq i$. Theoretically, negatives can be drawn from the entire dataset, i.e., any instance other than i . However, for computational efficiency, we approximate this by using other samples. Formally, the ranking scores for the modality $m \in \{eeg, face, eye\}$ are given by:

$$r_m^t = \langle \mathbf{z}_{m,i}^t, \mathbf{z}_{m,pos=i}^t \rangle - \langle \mathbf{z}_{m,i}^t, \mathbf{z}_{m,neg=j}^t \rangle, \quad j \neq i \quad (17)$$

$$r_m^s = \langle \mathbf{z}_{m,i}^s, \mathbf{z}_{m,pos=i}^s \rangle - \langle \mathbf{z}_{m,i}^s, \mathbf{z}_{m,neg=j}^s \rangle, \quad j \neq i \quad (18)$$

where $\langle \cdot, \cdot \rangle$ denotes the dot-product similarity. Instead of enforcing direct output consistency, the distillation objective focuses on preserving

relative similarity rankings between positive and negative instances. For each anchor sample, the student model is encouraged to maintain the same ordering of cross-modal similarities as learned by the unimodal teacher, which provides more stable supervision under heterogeneous feature distributions. The distillation loss for modality m is then formulated as:

$$\mathcal{L}_{SD}^m = \sum \max(0, r_m^t - r_m^s) \quad (19)$$

This hinge formulation ensures that the student achieves better pairwise ranking than the teacher, rather than merely mimicking it.

3.5.3. Generic distillation loss

To capture richer cross-instance relations, we complement the specific loss with a generic distillation loss, which aligns the softened score distributions between teacher and student. The loss is defined as:

$$\mathcal{L}_{GD}^m = - \sum [\tilde{r}_m^t \log \tilde{r}_m^s + (1 - \tilde{r}_m^t) \log (1 - \tilde{r}_m^s)] \quad (20)$$

$$\tilde{r}_m^t = \sigma\left(\frac{r_m^t}{\tau}\right) \quad (21)$$

$$\tilde{r}_m^s = \sigma\left(\frac{r_m^s}{\tau}\right) \quad (22)$$

where σ is the sigmoid function, τ is the softness of the temperature controlling the score, and $m \in \{eeg, face, eye\}$ denotes the modality. This loss transfers the dark knowledge of the teacher, encouraging the student to align with the global similarity structure throughout the dataset.

3.5.4. Counterfactual weighting

To balance the contributions of different modalities in multimodal distillation, we introduce counterfactual weighting. The core idea is to evaluate the relative importance of each modality in joint learning and adjust the distillation loss accordingly. The counterfactual weighting coefficients are derived by measuring the performance variation between the original multimodal prediction and its counterfactual counterparts, in which one modality is substituted by the teacher-guided representation of another. By quantifying how prediction confidence changes under such substitutions, the model can explicitly estimate the incremental contribution of each modality and dynamically adjust its influence during training. First, we compute joint and cross scores based on student and teacher representations:

$$\Delta s_i^m = r_m^s(i) - r_m^s(j) \quad (23)$$

$$\Delta t_i^m = r_m^t(i) - r_m^t(j) \quad (24)$$

$$\Delta s_i^{\bar{m}} = r_{\bar{m}}^s(i) - r_{\bar{m}}^s(j) \quad (25)$$

$$\Delta t_i^{\bar{m}} = r_{\bar{m}}^t(i) - r_{\bar{m}}^t(j) \quad (26)$$

$$J_i = (\Delta s_i^m \cdot \Delta s_i^{\bar{m}}) - (\Delta s_i^m \cdot \Delta s_j^{\bar{m}}) \quad (27)$$

where $\bar{m}$ is another modality. Then, we compute the counterfactual contribution from the joint scores, which reflects how under-optimized each modality is:

$$c_i^m = \frac{J_i - (\Delta s_i^{\bar{m}} \cdot \Delta t_i^m - \Delta s_i^{\bar{m}} \cdot \Delta t_j^m)}{(\Delta s_i^m \cdot \Delta t_i^{\bar{m}} - \Delta s_i^m \cdot \Delta t_j^{\bar{m}}) + \epsilon} \quad (28)$$

where ϵ is used to prevent division by zero in the formula. Finally, based on the computed counterfactual contributions, we derive the counterfactual weight for each modality, which adaptively amplifies the contribution of weaker modalities in distillation, and accordingly formulate the overall distillation objective as:

$$\lambda_i^m = 1 - \frac{c_i^m - c_i^{\bar{m}}}{c_i^m + c_i^{\bar{m}}} \quad (29)$$

$$\mathcal{L}_{dis}^m = \frac{1}{2} (\lambda_i^m \mathcal{L}_{SD}^m + \mathcal{L}_{GD}^m) \quad (30)$$

3.6. Data alignment

To achieve robust cross-modal semantic alignment while preserving intra-modal consistency under augmentation, we adopt a contrastive learning framework. Specifically, we introduce intra-modal and inter-modal contrastive losses to explicitly enforce that representations from the same instance remain similar under different augmentations, while different instances are pushed apart. This design ensures that the model learns meaningful semantic structures both within and across modalities. Unlike existing MER alignment methods that focus on a single representation view, our formulation jointly considers different augmentation strengths and cross-modal interactions, enabling multi-level and multi-view semantic alignment. The use of weak and strong augmentations is motivated by the need to improve model robustness. Weak augmentations preserve most of the original signal characteristics to stabilize learning, while strong augmentations encourage the network to focus on invariant semantic features, preventing overfitting to specific modalities or noise. By considering multiple semantic levels across both intra- and inter-modal representations, our framework ensures more comprehensive and robust alignment. Given weakly and strongly augmented embeddings of EEG and non-physiological modalities (facial expressions or eye movement), denoted as $\mathbf{Z}_e^w, \mathbf{Z}_e^s, \mathbf{Z}_f^w, \mathbf{Z}_f^s, \mathbf{Z}_o^w, \mathbf{Z}_o^s$, our objective enforces both intra-modal consistency and cross-modal semantic alignment.

3.6.1. Intra-modal contrastive loss

Within each modality, weak and strong views are encouraged to be similar, while other samples are treated as negatives. This is achieved using the normalized temperature-scaled cross-entropy (NT-Xent) loss (Liu et al., 2024):

$$\mathcal{L}_m^{intra} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\langle \mathbf{Z}_{m,i}^w, \mathbf{Z}_{m,i}^s \rangle / \tau)}{\sum_{j=1}^B \exp(\langle \mathbf{Z}_{m,i}^w, \mathbf{Z}_{m,j}^s \rangle / \tau)} \quad (31)$$

$$\mathcal{L}^{intra} = \mathcal{L}_e^{intra} + \mathcal{L}_f^{intra} \quad (32)$$

where $m \in \{eeg, face, eye\}$, B is the batch size.

3.6.2. Inter-modal alignment loss

To preserve cross-modal semantic consistency, we enforce alignment between EEG and facial embeddings under the same augmentation strength by minimizing their cosine distance:

$$\mathcal{L}_w^{align} = \frac{1}{B} \sum_{i=1}^B (1 - \cos(\mathbf{Z}_e^w, \mathbf{Z}_f^w)) \quad (33)$$

$$\mathcal{L}_s^{align} = \frac{1}{B} \sum_{i=1}^B (1 - \cos(\mathbf{Z}_e^s, \mathbf{Z}_f^s)) \quad (34)$$

$$\mathcal{L}_{aug}^{align} = \mathcal{L}_w^{align} + \mathcal{L}_s^{align} \quad (35)$$

3.6.3. Inter-modal contrastive loss

Additionally, a cross-modal NT-Xent Loss ensures discrimination between matched EEG-Face pairs and mismatched pairs:

$$\mathcal{L}_w^{inter} = \mathcal{L}_{NT-Xent}(\mathbf{Z}_e^w, \mathbf{Z}_f^w) \quad (36)$$

$$\mathcal{L}_s^{inter} = \mathcal{L}_{NT-Xent}(\mathbf{Z}_e^s, \mathbf{Z}_f^s) \quad (37)$$

$$\mathcal{L}^{inter} = \frac{1}{2} (\mathcal{L}_w^{inter} + \mathcal{L}_s^{inter}) \quad (38)$$

The final data alignment objective combines intra-modal consistency, inter-modal alignment, and inter-modal contrastive terms. This formulation ensures robust cross-modal semantic alignment while maintaining intra-modal consistency under data augmentation:

$$\mathcal{L}_{align} = \frac{1}{3} (\mathcal{L}^{intra} + \mathcal{L}^{align} + \mathcal{L}^{inter}) \quad (39)$$

3.7. Cross-modal attention fusion

We introduce a cross-modal attention interaction module to effectively integrate EEG signals with facial expression or eye movement features. This module establishes dynamic dependencies between heterogeneous modalities, enabling the model to capture complementary information across EEG and facial features.

$$\mathbf{F} = \text{Transformer}(\hat{\mathbf{H}}^e, \hat{\mathbf{H}}^f) \quad (40)$$

where $\hat{\mathbf{H}}^e$ denotes EEG feature representations, $\hat{\mathbf{H}}^f$ denotes facial or eye movement feature representations, and $\mathbf{F}$ is the fused cross-modal representation obtained through the Transformer block. The fused features are then passed into a classifier for prediction:

$$\hat{\mathbf{Y}} = \text{Classifier}(\mathbf{F}) \quad (41)$$

where $\hat{\mathbf{Y}}$ denotes the predicted class probabilities, and $\text{Classifier}(\cdot)$ is a fully connected classification network. For classification, we adopt the standard cross-entropy loss:

$$\mathcal{L}_{cls} = - \sum_{i=1}^N \mathbf{Y}_i \log \hat{\mathbf{Y}}_i \quad (42)$$

where $\mathbf{Y}_i$ is the ground-truth label for class, and $\hat{\mathbf{Y}}$ is the predicted probability for corresponding class. The overall training objective integrates classification loss, distillation loss, and contrastive loss:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{dis} + \lambda_2 \mathcal{L}_{align} \quad (43)$$

where λ_1, λ_2 are trade-off hyperparameters.

4. Experiment results and analysis

4.1. Emotion datasets

4.1.1. DEAP dataset

The DEAP dataset (Koelstra et al., 2011) contains EEG and peripheral physiological recordings from 32 participants while they watched 40 one-minute music video clips designed to elicit emotional responses. EEG signals were acquired using 32 electrodes in accordance with the international 10–20 system, and facial videos were recorded concurrently for all 22 participants. In our experiments, we used only the EEG data corresponding to those participants with available facial recordings to ensure consistent multimodal alignment. The EEG signals were preprocessed and band-pass filtered into four standard frequency bands: *theta* (4–8 Hz), *alpha* (8–13 Hz), *beta* (13–30 Hz), and *gamma* (30–45 Hz). Then, DE is extracted as a shallow representation for each 1-s segment of the signal with no overlap.

4.1.2. SEED-IV dataset

The SEED-IV dataset (Zheng & Lu, 2015), an extension of the SEED dataset for multi-class emotion recognition using EEG signals, was developed by the BCMI Laboratory at Shanghai Jiao Tong University. The dataset contains EEG recordings from 15 participants while they watched 72 film clips selected to elicit four discrete emotional states: happiness, sadness, fear, and neutrality. EEG signals were recorded using 62 channels based on the international 10–20 system at a sampling rate of 1000 Hz and later downsampled to 200 Hz. Each participant participated in three sessions on separate days to evaluate the stability of emotional EEG features within and across sessions. The SEED-IV dataset is widely used for emotion recognition, domain adaptation, and multimodal fusion research due to its high-quality labeling, multi-session design, and well-controlled experimental protocol.

4.2. Experimental setting

The input features have dimensions of $880 \times 32 \times 60 \times 4$ for EEG, $880 \times 32 \times 11$ for facial expressions, and $880 \times 31 \times 10$ for eye movements. To balance the contributions of the counterfactual knowledge

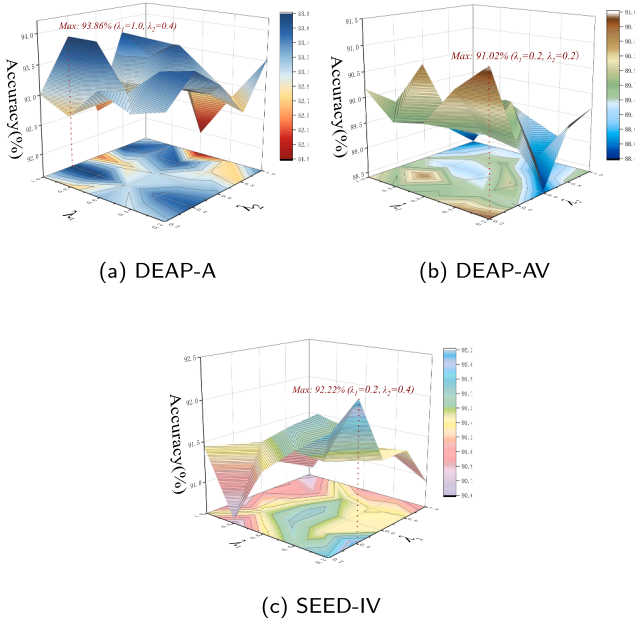


Fig. 2. The results of hyperparameter λ_1 and λ_2 in the two dataset.

distillation loss and the alignment loss, two hyperparameters, λ_1 and λ_2 , are introduced in the loss function. A systematic parameter search is conducted for both λ_1 and λ_2 over the range 0.2, 0.4, 0.6, 0.8, 1.0 using five-fold cross-validation. As shown in Fig. 2, different combinations of λ_1 and λ_2 lead to varying performance. The optimal parameter settings are $\lambda_1 = 1.0$ and $\lambda_2 = 0.4$ for the DEAP-A dataset, $\lambda_1 = 0.2$ and $\lambda_2 = 0.2$ for the DEAP-AV dataset, and $\lambda_1 = 0.2$ and $\lambda_2 = 0.4$ for the SEED-IV dataset.

To comprehensively evaluate the performance and generalization capability of the proposed model, a five-fold cross-validation strategy is adopted. The dataset is randomly partitioned into five equal subsets, with four subsets used for training and the remaining subset used for testing in each fold, and the final performance is reported as the average accuracy across all folds. The proposed MBDA framework is implemented in Python using the PyTorch library and trained on a single NVIDIA GPU. The Adam optimizer is employed with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-3} . The batch size is set to 32, and all models are trained for 500 epochs. To stabilize training and alleviate overfitting, gradient clipping with a maximum norm of 5.0 and a dropout rate of 0.1 are applied. Unimodal teacher models are pre-trained independently using single-modality inputs and remain frozen during multimodal training, while the multimodal student is trained jointly with task supervision and the proposed counterfactual distillation objective. All experiments are conducted with a fixed random seed of 1024 to ensure reproducibility.

4.3. Comparison with state-of-the-art methods

To comprehensively validate the effectiveness and superiority of the proposed MBDA framework, we conducted extensive comparative experiments with several state-of-the-art MER approaches on two public benchmark datasets. As summarized in Table 1, MBDA consistently achieves the highest performance across all datasets, establishing new benchmarks for multimodal emotion analysis.

Specifically, on the DEAP-A dataset, MBDA attains an accuracy of 93.86% and an F1-score of 93.20%, surpassing GWO (92.50%) (Singh et al., 2023), DISD-Net (75.00%) (Cheng et al., 2025) and DG-JCA (92.89%) (Cheng et al., 2024). On DEAP-V, MBDA reaches 95.11% accuracy and 95.00% F1-score, outperforming DG-JCA (92.50%) (Cheng et al., 2024) and DISD-Net (73.07%) (Cheng et al., 2025), highlighting its enhanced cross-modal discriminative ability. For the DEAP-AV

dataset, where only AVE-LSTM has reported results (69.3%) (Chen et al., 2022), MBDA achieves 91.02% accuracy and 89.20% F1-score, demonstrating substantial improvement. Finally, on SEED-IV, MBDA attains 92.66% accuracy and 90.80% F1-score, outperforming MFFNN (87.32% accuracy / 85.04% F1-score) (Fu et al., 2023), SOFNN (86.27% accuracy / 80.83% F1-score) (Zhou et al., 2022), GMHCA (86.79% accuracy / 86.13% F1-score) (Li et al., 2025), CLDTA (88.30%) (Liao et al., 2024), DG-JCA (88.24%) (Cheng et al., 2024), DISD-Net (65.65%) (Cheng et al., 2025) and SS-EMERGE (81.51%) (Ahuja & Sethia, 2025), indicating that MBDA effectively captures complementary information across modalities and achieves robust and balanced multimodal emotion recognition.

In summary, MBDA demonstrates consistent superiority over existing MER methods by synergistically integrating modality-aware data augmentation, multi-level feature alignment, and counterfactual knowledge distillation. This coordinated design allows the model to effectively leverage complementary emotional cues, enhance the representation of weaker modalities, and maintain cross-modal consistency even under variable conditions. As a result, MBDA achieves higher accuracy, improved robustness, and strong generalization. These findings underscore the effectiveness of the proposed framework and establish it as a reliable and generalizable paradigm for multimodal emotion recognition in both controlled and real-world scenarios.

4.4. Ablation study

To verify the effectiveness of each component in the proposed model, a series of ablation experiments is conducted. The analysis focuses on four aspects: loss functions, fusion strategies, enhancement methods, and modality. These experiments elucidate the contribution of each module and demonstrate their combined effect on enhancing the overall performance of MER.

4.4.1. Loss function ablation

To verify the overall performance of the proposed model and the effectiveness of each loss component, we conducted an ablation study on the loss functions. The total loss consists of three parts: classification loss, alignment loss, and counterfactual knowledge distillation loss. A series of comparative experiments was performed by systematically omitting each loss term to evaluate its contribution to the model's performance.

As illustrated in the Table 2, training with only the classification loss yields the lowest accuracy across all datasets, indicating that a single objective cannot capture complex inter-modal relationships. Adding the alignment loss improves performance by enforcing cross-modal consistency, but the improvement is limited as modality imbalance remains unaddressed. Incorporating counterfactual knowledge distillation further enhances performance by adaptively balancing modality contributions, strengthening weaker modalities, and mitigating dominance from stronger ones.

However, when classification, alignment, and counterfactual distillation losses are jointly optimized, the model achieves the best performance, with accuracies of 93.86%, 95.11%, 91.0%, and 92.66% and corresponding F1-scores of 93.20%, 95.00%, 89.20%, and 90.80% on the DEAP-A, DEAP-V, DEAP-AV, and SEED-IV datasets, respectively. This demonstrates that the three objectives play complementary roles: the classification loss ensures task discriminability, the alignment loss promotes semantic coherence, and the knowledge distillation loss enhances modality balance. Their synergistic integration yields a more robust and generalizable multimodal representation, confirming the effectiveness of the proposed loss design.

4.4.2. Augment method ablation

To further assess the effectiveness of the proposed data augmentation strategy, we conducted ablation experiments under varying augmentation settings. As illustrated in Fig. 3, three configurations were compared: weak enhancement, strong enhancement, and the combination

Table 1
Comparison with accuracies and F1-score on the DEAP, SEED-IV, and SEED-V datasets.

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV
MFFNN (Fu et al., 2023)	–	–	–	87.32 / 85.04
SS-EMERGE (Ahuja & Sethia, 2025)	–	–	–	81.51 / –
SOFNN (Zhou et al., 2022)	–	–	–	86.27 / 80.83
CLDTA (Liao et al., 2024)	–	–	–	88.30 / –
GWO (Singh et al., 2023)	92.50 / –	81.25 / –	–	–
DG-JCA (Cheng et al., 2024)	92.89 / –	92.50 / –	–	88.24 / –
DISD-Net (Cheng et al., 2025)	75.00 / –	73.07 / –	–	65.65 / –
AVE-LSTM (Chen et al., 2022)	–	–	69.3 / –	–
GMHCA (Li et al., 2025)	–	–	–	86.79 / 86.13
Ours	93.86 / 93.20	95.11 / 95.00	91.02 / 89.20	92.66 / 90.80

Note: Values are reported as Accuracy (%) / F1-score (%).

Table 2
The ablation results of loss on DEAP and SEED-IV datasets.

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV
$\mathcal{L}_{cls}$	91.81 / 89.70	93.35 / 91.20	95.45 / 93.17	91.06 / 89.03
$\mathcal{L}_{cls} + \mathcal{L}_{dis}$	92.72 / 90.43	94.66 / 92.77	90.34 / 87.98	91.16 / 89.84
$\mathcal{L}_{cls} + \mathcal{L}_{align}$	92.67 / 91.03	94.53 / 93.13	89.94 / 88.07	92.54 / 90.37
$\mathcal{L}_{cls} + \mathcal{L}_{dis} + \mathcal{L}_{align}$	93.86 / 93.20	95.11 / 95.00	91.02 / 89.20	92.66 / 90.80

Note: Values are reported as Accuracy (%) / F1-score (%).

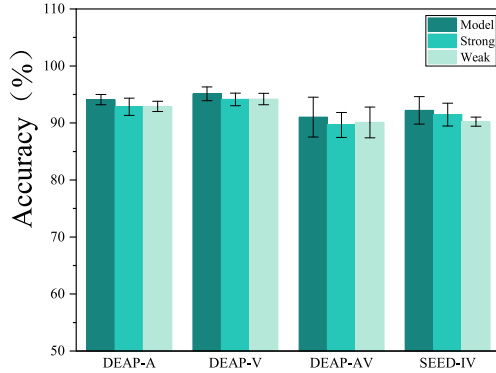


Fig. 3. The performance of augment on the DEAP-A, DEAP-V, DEAP-AV, and SEED-IV datasets.

of both. The results demonstrate that applying a single enhancement strategy brings only limited improvement, whereas the combination of strong and weak enhancement achieves the best and most stable performance across all datasets.

The results on the DEAP-A, DEAP-V, DEAP-AV, and SEED-IV datasets demonstrate that the model incorporating both strong and weak augmentations achieves consistently high accuracy, typically around or above 90%, highlighting the effectiveness of integrating these augmentation strategies into the base model. Specifically, on the DEAP-A, DEAP-V, and SEED-IV datasets, the combined enhancement strategy significantly outperforms the single-method results, demonstrating the complementary benefits of the two augmentation approaches. Although the accuracy on the DEAP-AV dataset is relatively lower due to its higher complexity, the combined method still surpasses the single-enhancement baselines, showing good robustness and stability. Overall, the combined use of strong and weak augmentations effectively enhances the model's performance and generalization across datasets.

4.4.3. Fusion methods ablation

To further evaluate the effectiveness of the cross-fusion strategy proposed for multimodal feature integration, we performed comparative experiments with two baseline fusion methods: element-wise addition (Add) and feature concatenation (Cat). Each fusion strategy was applied to modality-specific features extracted by the respective encoders,

and the fused representations were then used for classification. The performance comparison was evaluated through analysis of receiver operating characteristic (ROC) curves and the associated area under the curve (AUC) values across four datasets: DEAP-V, SEED-IV, DEAP-AV, and DEAP-A. The results are illustrated in Fig. 4.

As shown in the figure, the performance trends across the four datasets exhibit high consistency. The proposed Cross fusion method consistently achieves the best classification performance among all three fusion approaches. For instance, on the SEED-IV dataset, Cross attains an AUC of 95.6%, outperforming both Add (91.0%) and Cat (94.6%). Similar improvements are observed on other datasets, such as an AUC of 95.1% for Cross compared to 93.7% for Add and 94.3% for Cat. These results demonstrate that the Cross-Fusion approach effectively integrates multimodal information and captures complex inter-modal relationships, thereby enhancing classification performance. Moreover, the relatively lower AUC scores of the Add and Cat fusion methods highlight their limitations. The simple addition operation may cause feature information loss, while direct concatenation—though better—still lacks sufficient interaction between modalities. In contrast, the Cross fusion strategy allows richer cross-modal feature exchange and better alignment, thereby enhancing the model's discriminative capacity.

In summary, across all datasets, the Cross fusion method consistently yields the highest AUC values and the most stable performance, demonstrating its effectiveness and superiority in multimodal feature integration for emotion recognition tasks.

4.4.4. The performance of modality

To assess the contribution of each modality, we performed a modality ablation study by individually evaluating the classification performance of each unimodal input and comparing the results with those obtained from the fused multimodal representation (see Fig. 5). This experiment enables us to assess the relative effectiveness of individual modalities and the overall benefits of multimodal integration.

On the DEAP dataset, both EEG and facial modalities achieved lower accuracy when used independently, indicating that each modality alone provides limited emotional representation. However, when the two modalities were fused, the classification accuracy significantly improved, demonstrating that the integration of EEG and facial features effectively combines physiological and expressive cues to enhance emotion recognition performance. A similar pattern was observed on the SEED-IV dataset, where EEG and eye-movement modalities each contributed partial but insufficient discriminative information. The fusion

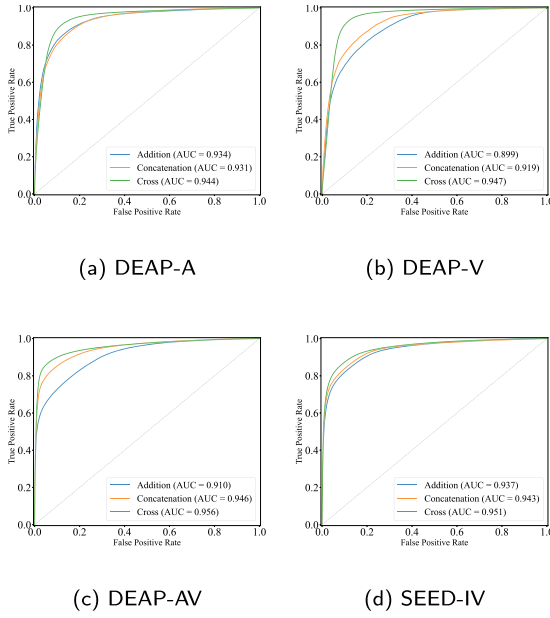


Fig. 4. The ROC curves of different fusion methods on DEAP and SEED-IV datasets.

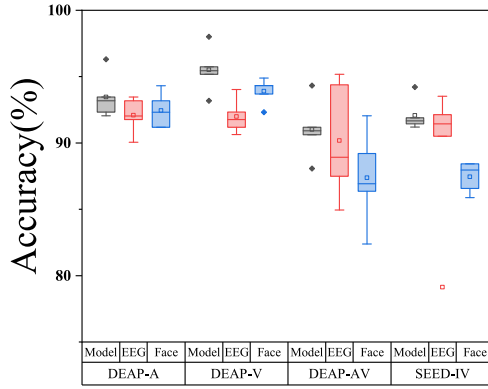


Fig. 5. The performance of modality on the DEAP-A, DEAP-V, DEAP-AV, and SEED-IV datasets.

of the two modalities consistently yielded superior results, confirming that multimodal learning allows the model to leverage complementary information from different signal sources, thereby improving robustness and generalization.

Overall, the ablation results clearly show that unimodal features alone are insufficient for robust emotion recognition, while multimodal fusion significantly enhances performance across all datasets. These findings highlight the necessity of integrating multiple modalities to capture richer and more comprehensive emotional representations.

4.5. Visualization analysis

4.5.1. Feature distribution

To assess the contribution of each module, the learned feature distributions were visualized using t-SNE under four experimental settings: (1) classification loss only, (2) classification with counterfactual knowledge distillation, (3) classification with contrastive learning, and (4) the complete model. As shown in Fig. 6, using only the classification loss results in overlapping features with weak separability, indicating limited discrimination. Adding contrastive learning improves class boundaries and intra-class compactness, enhancing feature discrimination. Introducing counterfactual knowledge distillation further structures the fea-

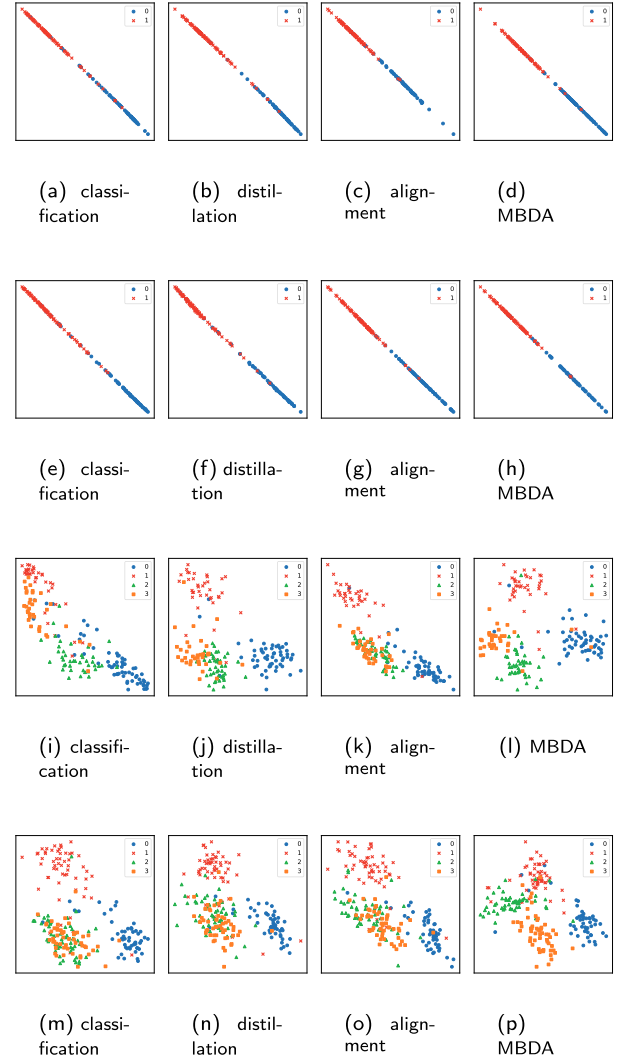


Fig. 6. t-SNE Visualization of Feature Distributions under four Settings: Classification, Classification With Distillation, Classification With Alignment, and Full Model (Rows 1–4 correspond to DEAP-A, DEAP-V, DEAP-AV, SEED-IV).

ture space with well-separated clusters, facilitating cross-modal knowledge transfer and robustness. Overall, the full model produces the most discriminative and separable features, demonstrating the complementary effectiveness of all components.

4.5.2. Confusion matrix analysis

To further understand the model's performance, we analyzed the confusion matrices shown in Fig. 7 for both the DEAP and SEED-IV datasets. For the valence dimension of DEAP, positive and negative emotions achieve accuracies of 92.88% and 97.35%, respectively, while for arousal, positive and negative emotions reach 93.84% and 92.55%. These results indicate that the model maintains balanced sensitivity to different affective polarities, with slightly better recognition of positive emotional states. This may be due to EEG and facial features providing complementary neural and behavioral cues, allowing the model to capture subtle variations in emotional responses.

In the four-class DEAP task, the model achieves 89.12%, 87.84%, 90.64%, and 89.21%, respectively. Misclassifications mainly occur between adjacent categories such as 'Happy-Angry' and 'Sad-Relaxed', likely because these emotions share overlapping affective dimensions (e.g., arousal or valence), leading to similar spatiotemporal EEG patterns and facial expressions. For SEED-IV, the model attains 92.96% (Neutral), 90.19% (Sad), 85.37% (Fear), and 94.63% (Happy). The high accuracy

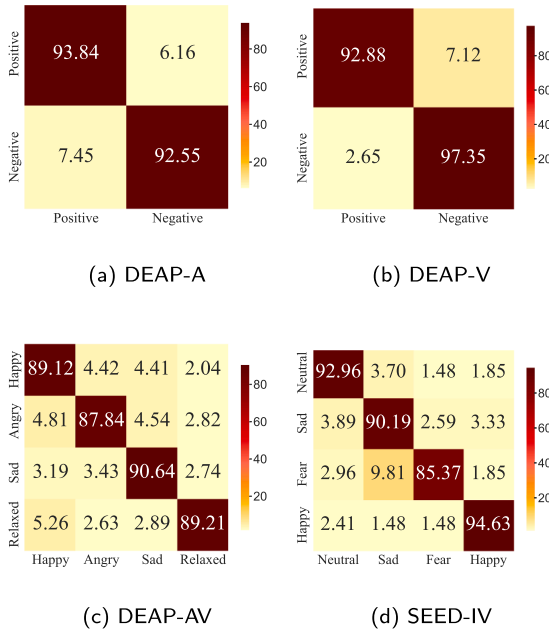


Fig. 7. Confusion matrices of MBDA on DEAP and SEED-IV datasets.

for Happy suggests strong intra-class consistency, whereas the lower accuracy for Sad reflects potential confusion with other negative emotions, likely due to shared cortical activation patterns and facial indicators.

These results show that the model performs stably and discriminatively across both binary and multi-class emotion recognition tasks. The joint modeling of multimodal features, along with intra- and inter-modal attention, enables the model to capture distinctive patterns for each emotion, supporting robust recognition across datasets.

4.5.3. Dynamic modality weights

As shown in Fig. 8, we visualize the dynamic evolution of EEG and Face modality weights over 500 training epochs. It can be observed that the weights of both modalities fluctuate within a limited range and do not converge to fixed values, indicating that the model continuously adjusts modality importance during training. In some experimental settings, the Face modality exhibits slightly higher weights, while in others the two modalities alternate in dominance, reflecting adaptive modality prioritization. Moreover, frequent weight crossings suggest that EEG and Face provide complementary emotional cues rather than redundant information. Overall, the dynamic and balanced weight distributions demonstrate the effectiveness of the proposed modality-balancing strategy in enabling flexible and robust multimodal fusion.

4.5.4. Stability and convergence analysis of accuracy curves

Fig. 9 illustrates the test accuracy curves of the four experimental settings over 500 training epochs for SEED-IV, DEAP-A, DEAP-AV, and DEAP-V. For DEAP-A, accuracy rises steadily, reaching around 0.88 within the first 50 epochs and stabilizing near 0.91, with minor fluctuations, indicating strong generalization. For SEED-IV, convergence is slightly slower, taking about 100–150 epochs to reach 0.88–0.90, with larger early-stage oscillations. DEAP-AV shows a rapid increase, reaching 0.88 within 40 epochs and maintaining stable performance thereafter. DEAP-V demonstrates the fastest convergence, surpassing 0.88 within 30–40 epochs and stabilizing around 0.91. Across all datasets, final accuracies are consistently high (0.88–0.91), and no overfitting is observed. These results indicate that MBDA achieves stable and robust multimodal emotion recognition, with convergence speed and early-stage stability influenced by dataset characteristics and hyperparameter settings.

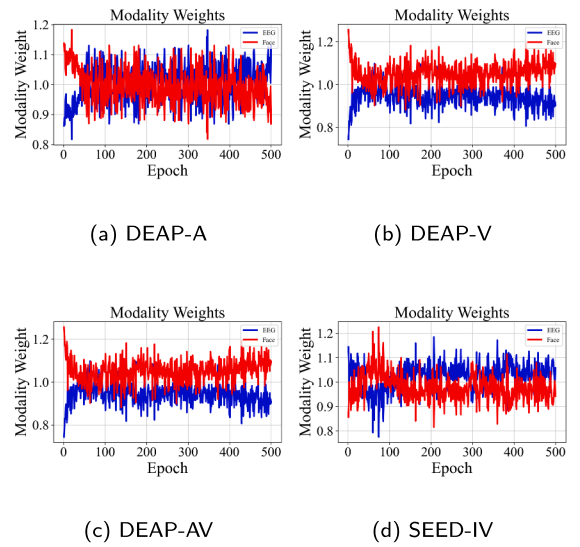


Fig. 8. Dynamic Modality Weights of MBDA on DEAP and SEED-IV datasets.

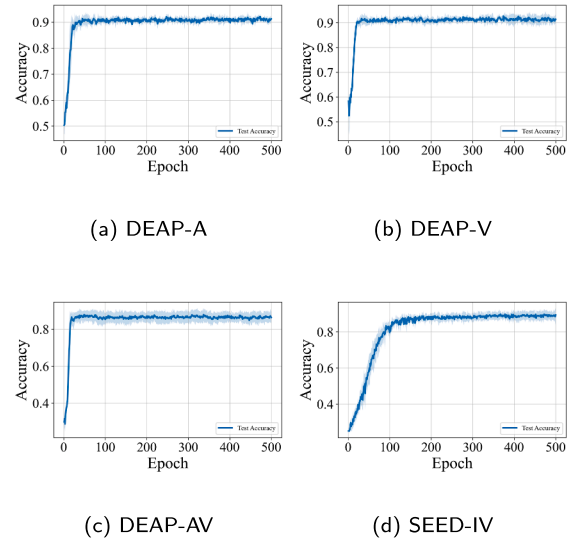


Fig. 9. Accuracy Curves of each epoch on DEAP and SEED-IV datasets.

5. Discussion

In this study, the proposed MBDA framework significantly improves MER accuracy, particularly in the fusion of EEG, facial expressions, and eye movement data. However, despite the promising results, a deeper analysis is necessary to identify potential areas of improvement and limitations of the proposed method.

5.1. Performance analysis and diagnostic interpretation

The proposed MBDA framework demonstrates consistent performance improvements across the DEAP and SEED-IV datasets compared with existing unimodal and bimodal methods. The comparison results and ablation studies indicate that the performance gains are not solely due to multimodal fusion, but are jointly attributed to modality-balanced training, data augmentation, and multi-level alignment. In particular, removing either the alignment strategy or the distillation mechanism leads to noticeable performance degradation, highlighting their complementary roles in improving robustness and generalization.

Further diagnostic analysis based on confusion matrices provides insight into emotion-specific sensitivity. On the DEAP dataset, MBDA

achieves balanced recognition performance on both arousal and valence dimensions, while in the four-class task, most misclassifications occur between emotionally adjacent categories such as Happy–Angry and Sad–Relaxed. This observation is consistent with prior studies and reflects intrinsic overlaps in affective representations rather than model instability. On SEED-IV, the model shows relatively stronger discrimination for emotions such as Happy and Neutral, while emotions like Sad and Fear remain more challenging due to subtle and overlapping physiological patterns, which is also commonly reported in existing literature.

Overall, these results suggest that MBDA is particularly effective in capturing discriminative multimodal patterns while maintaining stable decision boundaries. Importantly, the interpretability of MBDA arises from its training dynamics rather than post-hoc analysis: the modality balancing and alignment mechanisms provide diagnostic insight into how different modalities contribute to emotion recognition, supporting both performance improvement and model transparency.

5.2. Findings and limitations

Multimodal emotion recognition often suffers from inherent imbalances and distribution shifts across modalities, particularly when weaker modalities are more sensitive to perturbations. Strong augmentations can exacerbate these shifts, causing semantic drift and instability in weaker modalities, which in turn makes them more easily dominated during fusion. The MBDA framework addresses this challenge by stabilizing modality-level representations under varying augmentation strengths, ensuring that informative patterns from weaker modalities are consistently preserved and utilized. Furthermore, the multi-level alignment strategy mitigates augmentation-induced discrepancies both within and across modalities, producing more coherent and reliable multimodal representations.

Although the MBDA framework achieves promising results, several limitations should be acknowledged. First, the model's reliance on robust multimodal data means that its performance may degrade in the presence of noise or insufficient data from certain modalities. For instance, when weaker modalities (such as facial expressions or eye movement) provide less discriminative information, the model's overall recognition ability might be limited, even with modality balancing. Second, the current framework assumes that augmentations, both weak and strong, equally benefit all modalities. However, in practice, the effectiveness of different augmentation strengths may vary across modalities, potentially leading to suboptimal performance in some cases. Third, while the alignment strategy effectively enhances cross-modal consistency, it may not fully address the challenge of aligning features from highly heterogeneous modalities when faced with complex or highly variable emotion expressions. Lastly, the MBDA framework's performance can vary depending on certain design choices, which may affect its stability and generalizability across different datasets.

6. Conclusion

This paper presents MBDA, a unified Modality-Balanced framework with Data Augmentation and Alignment for MER. By addressing challenges such as modality imbalance, cross-modal inconsistency, and limited data diversity, MBDA enhances model performance through data augmentation, multi-level feature alignment, and counterfactual knowledge distillation. The combination of weak and strong augmentations increases data diversity, while feature alignment ensures cross-modal consistency. Counterfactual distillation prevents dominant modalities from overshadowing weaker ones, improving robustness. Experiments on DEAP and SEED-IV show that MBDA consistently outperforms state-of-the-art methods, achieving competitive accuracies. Future work will focus on extending MBDA to dynamic modality configurations and incorporating additional modalities, enhancing its applicability in real-world scenarios.

CRedit authorship contribution statement

Cheng Cheng: Writing – original draft, Methodology, Funding acquisition, Conceptualization; **Ruisi Shang:** Visualization, Methodology, Investigation; **Zixu Wang:** Conceptualization, Visualization, Methodology; **Huazhi Li:** Writing – review & editing, Visualization, Methodology; **Ziyu Jia:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Ahuja, C., & Sethia, D. (2025). SS-EMERGE-self-supervised enhancement for multidimensional emotion recognition using GNNs for EEG. *Scientific Reports*, 15(1), 14254.
- Baek, J.-Y., & Lee, S.-P. (2023). Enhanced speech emotion recognition using DCGAN-based data augmentation. *Electronics*, 12(18), 3966.
- Bao, G., Yan, B., Tong, L., Shu, J., Wang, L., Yang, K., & Zeng, Y. (2021). Data augmentation for EEG-based emotion recognition using generative adversarial networks. *Frontiers in Computational Neuroscience*, 15, 723843.
- Cai, Y., Ye, R., Xie, J., Zhou, Y., Xu, Y., & Wu, Z. (2024). Robust representation learning for multimodal emotion recognition with contrastive learning and mixup. In *Proceedings of the 2nd international workshop on multimodal and responsible affective computing* (pp. 93–97).
- Chen, C., & Zhang, P. (2024). Modality-collaborative transformer with hybrid feature reconstruction for robust emotion recognition. *ACM Transactions on Multimedia Computing, Communications and Applications*, 20(5), 1–23.
- Chen, J., Ro, T., & Zhu, Z. (2022). Emotion recognition with audio, video, EEG, and EMG: A dataset and baseline approaches. *IEEE Access*, 10, 13229–13242.
- Cheng, C., Liu, W., Feng, L., & Jia, Z. (2024). Dense graph convolutional with joint cross-attention network for multimodal emotion recognition. *IEEE Transactions on Computational Social Systems*, 11(5), 6672–6683.
- Cheng, C., Liu, W., Wang, X., Feng, L., & Jia, Z. (2025). DISD-Net: A dynamic interactive network with self-distillation for cross-subject multi-modal emotion recognition. *IEEE Transactions on Multimedia*, 27, 4643–4655.
- Fu, B., Gu, C., Fu, M., Xia, Y., & Liu, Y. (2023). A novel feature fusion network for multimodal emotion recognition from EEG and eye movement signals. *Frontiers in Neuroscience*, 17, 1234162.
- Guo, Y., Zhang, T., & Huang, W. (2023). Emotion recognition based on multi-modal electrophysiology multi-head attention contrastive learning. *arXiv:2308.01919*.
- Jiang, Q., Huang, L., & Yang, Y. (2025). Rethinking multimodal learning from the perspective of mitigating classification ability disproportion. *arXiv:2502.20120*.
- Jin, Y., Liu, X., Yang, Y., Yu, Z., Zhang, T., & Yang, K. (2025). RoHyDR: Robust hybrid diffusion recovery for incomplete multimodal emotion recognition. *arXiv:2505.17501*.
- Kim, J., Hong, J., & Choi, Y. (2024). Causal inference for modality debiasing in multimodal emotion recognition. *Applied Sciences*, 14(23), 11397.
- Koelstra, S., Muhl, C., Soleymani, M., Lee, J.-S., Yazdani, A., Ebrahimi, T., Pun, T., Nijholt, A., & Patras, I. (2011). Deap: A database for emotion analysis; using physiological signals. *IEEE Transactions on Affective Computing*, 3(1), 18–31.
- Li, D., Wang, Y., Funakoshi, K., & Okumura, M. (2023a). Joyful: Joint modality fusion and graph contrastive learning for multimodal emotion recognition. *arXiv:2311.11009*.
- Li, Q., Gao, Y., Wen, Y., Wang, C., & Li, Y. (2024). Enhancing modal fusion by alignment and label matching for multimodal emotion recognition. *arXiv:2408.09438*.
- Li, X., Li, Y., Li, Y., & Yang, Y. (2025). GMHCA-MCBILSTM: A gated multi-head cross-modal attention-based network for emotion recognition using multi-physiological signals. *Algorithms*, 18(10), 664.
- Li, Y., Wang, Y., & Cui, Z. (2023b). Decoupled multimodal distilling for emotion recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 6631–6640).
- Lian, H., Lu, C., Li, S., Zhao, Y., Tang, C., & Zong, Y. (2023). A survey of deep learning-based multimodal emotion recognition: Speech, text, and face. *Entropy*, 25(10), 1440.
- Liao, Y., Zhang, Y., Wang, S., Zhang, X., Zhang, Y., Chen, W., Gu, Y., & Huang, L. (2024). CLDTA: Contrastive learning based on diagonal transformer autoencoder for cross-dataset EEG emotion recognition. *arXiv:2406.08081*.
- Liu, Q., Hao, J., & Guo, Y. (2023). EEG data augmentation for emotion recognition with a task-driven GAN. *Algorithms*, 16(2), 118.

- Liu, Q., Zhou, Z., Wang, J., & Liang, Z. (2024). Joint contrastive learning with feature alignment for cross-corpus EEG-based emotion recognition. In *Proceedings of the 1st international workshop on brain-computer interfaces (BCI) for multimedia understanding* (pp. 9–17).
- Ma, X., Chen, H., & Deng, Y. (2025). Improving multimodal learning balance and sufficiency through data remixing. arXiv:2506.11550.
- Nguyen, C.-V. T., Le, T.-S., Mai, A.-T., & Le, D.-T. (2024). Ada2I: Enhancing modality balance for multimodal conversational emotion recognition. In *Proceedings of the 32nd ACM international conference on multimedia* (pp. 9330–9339).
- Roh, K.-M., & Lee, S.-P. (2024). Enhanced speech emotion recognition using conditional-dcgan-based data augmentation. *Applied Sciences*, 14(21), 9890.
- Roy, A. K., Kathania, H. K., & Sharma, A. (2024). Improvement in facial emotion recognition using synthetic data generated by diffusion model. arXiv:2411.10863.
- Shi, T., & Huang, S.-L. (2023). MultiEMO: An attention-based correlation-aware multimodal fusion framework for emotion recognition in conversations. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 14752–14766).
- Shou, Y., Meng, T., Ai, W., Yin, N., & Li, K. (2023). Adversarial representation with intra-modal and inter-modal graph contrastive learning for multimodal emotion recognition. arXiv:2312.16778.
- Siddhad, G., Iwamura, M., & Roy, P. P. (2024). Enhancing EEG signal-based emotion recognition with synthetic data: Diffusion model approach. arXiv:2401.16878.
- Singh, U., Shaw, R., & Patra, B. K. (2023). A data augmentation and channel selection technique for grading human emotions on DEAP dataset. *Biomedical Signal Processing and Control*, 79, 104060.
- Vairamani, A. D. (2024). Advancements in multimodal emotion recognition: integrating facial expressions and physiological signals. In *Affective computing for social good: Enhancing well-being, empathy, and equity* (pp. 217–240). Springer.
- Wang, X., Zhao, S., Sun, H., Wang, H., Zhou, J., & Qin, Y. (2025). Enhancing multimodal emotion recognition through multi-granularity cross-modal alignment. In *ICASSP 2025-2025 IEEE international conference on acoustics, speech and signal processing (ICASSP)* (pp. 1–5). IEEE.
- Wang, Y., Li, Y., Liang, P. P., Morency, L.-P., Bell, P., & Lai, C. (2023). Cross-attention is not enough: Incongruity-aware dynamic hierarchical fusion for multimodal affect recognition. arXiv:2305.13583.
- Wang, Z., Li, D., Li, G., Zhang, Z., & Jiang, R. (2024). Multimodal low-light image enhancement with depth information. In *Proceedings of the 32nd ACM international conference on multimedia* (pp. 4976–4985).
- Woo, S., Zubair, M., Lim, S., & Kim, D. (2025). Deep multimodal emotion recognition using modality-aware attention and proxy-based multimodal loss. *Internet of Things*, 31, 101562.
- Wu, S., Liang, H., Zhang, Y., Chen, Y., & Jia, Z. (2025). A cross-modal densely guided knowledge distillation based on modality rebalancing strategy for enhanced unimodal emotion recognition. In *Proceedings of the thirty-fourth international joint conference on artificial intelligence, IJCAI 2025, Montreal, Canada, 16–22, 2025* (pp. 4236–4244).
- Wu, Y., Mi, Q., & Gao, T. (2025). A comprehensive review of multimodal emotion recognition: Techniques, challenges, and future directions. *Biomimetics*, 10(7), 418.
- Xie, Y., Sun, C.-J., Cao, Z., Liu, B., Ji, Z., Liu, Y., & Shan, L. (2025). A dual contrastive learning framework for enhanced multimodal conversational emotion recognition. In *Proceedings of the 31st international conference on computational linguistics* (pp. 4055–4065).
- Zhang, J., Liu, G., Liu, Q., Wu, S., & Wang, L. (2024). Modality-balanced learning for multimedia recommendation. In *Proceedings of the 32nd ACM international conference on multimedia* (pp. 7551–7560).
- Zhao, Z., Chen, H., Li, X., Jiang, D., & Xie, L. (2024). Improving multimodal emotion recognition by leveraging acoustic adaptation and visual alignment. In *Proceedings of the 2nd international workshop on multimodal and responsible affective computing* (pp. 67–71).
- Zheng, W.-L., & Lu, B.-L. (2015). Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks. *IEEE Transactions on Autonomous Mental Development*, 7(3), 162–175.
- Zhou, S., Huang, D., Liu, C., & Jiang, D. (2022). Objectivity meets subjectivity: A subjective and objective feature fused neural network for emotion recognition. *Applied Soft Computing*, 122, 108889.